

STILLWATER, OK, USA

WMO: 723545

Lat: 36.162N

Lon: 97.089W

Elev: 984

StdP: 14.18

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 03965

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.3	17.8	1.6	6.2	18.3	6.7	8.0	22.6	26.3	50.8	24.4	49.8	7.2	0	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	102.0	74.6	99.1	75.1	95.6	75.3	78.9	93.3	77.8	92.1	76.8	90.7	11.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.0	136.1	86.2	73.4	129.0	84.3	72.7	125.8	83.5	43.1	93.4	41.9	92.0	40.8	90.9	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.7	21.9	19.6	DB	3.9	105.2	6.1	4.3	-0.5	108.2	-4.1	110.8	-7.5	113.2	-12.0	116.3	(4)
(5)			WB	3.4	81.3	5.9	2.4	-0.8	83.0	-4.3	84.4	-7.6	85.7	-11.9	87.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.9	38.3	42.1	51.3	60.3	69.0	77.9	82.9	81.9	74.2	61.7	50.1	39.9	(6)	
	DBStd	18.02	10.51	10.85	11.05	9.29	7.72	5.66	5.44	5.75	7.46	9.20	9.66	10.02	(7)	
	HDD50	1250	383	263	117	21	1	0	0	0	0	14	116	335	(8)	
	HDD65	3576	827	644	441	197	50	1	0	0	13	173	453	777	(9)	
	CDD50	5229	21	41	157	329	589	836	1021	990	725	378	119	23	(10)	
	CDD65	2079	0	1	16	54	173	387	556	525	288	72	7	0	(11)	
	CDH74	24087	5	20	163	586	1677	4333	7054	6364	3075	731	77	2	(12)	
	CDH80	12056	0	3	33	164	590	2042	3998	3545	1441	230	10	0	(13)	
(14) Wind	WSAvg	9.4	9.1	9.7	10.9	11.4	10.1	9.9	8.5	8.1	8.3	9.0	9.2	8.9	(14)	
(15) Precipitation	PrecAvg	34.6	1.2	1.4	2.7	3.1	4.9	4.3	3.0	3.1	4.1	3.0	2.3	1.6	(15)	
	PrecMax	46.4	3.7	5.4	7.7	8.3	14.6	10.6	8.2	8.5	12.4	8.7	6.7	5.1	(16)	
	PrecMin	19.7	0.0	0.1	0.0	0.2	0.6	0.8	0.0	0.0	0.0	0.4	0.0	0.1	(17)	
	PrecStd	6.7	1.1	1.2	1.9	1.9	2.8	2.4	1.9	2.1	2.6	2.1	1.9	1.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.5	76.6	83.8	90.2	93.1	100.1	106.4	107.7	102.1	90.4	81.2	72.2	(19)	
		MCWB	53.7	58.8	62.8	67.0	71.5	75.3	73.9	73.2	73.9	70.3	62.5	57.5	(20)	
	2%	DB	66.4	71.4	78.9	84.2	89.9	96.5	102.2	102.1	94.9	86.0	75.3	65.8	(21)	
		MCWB	51.2	55.3	60.2	64.7	71.7	74.8	74.4	74.9	73.5	67.8	59.8	55.5	(22)	
	5%	DB	61.4	65.7	73.4	81.0	86.4	92.8	99.1	99.1	91.4	81.9	71.6	61.1	(23)	
		MCWB	49.1	54.2	58.7	63.8	70.4	74.9	75.5	75.1	72.8	65.6	58.4	50.7	(24)	
	10%	DB	55.4	61.0	69.8	76.7	82.6	90.3	95.7	95.2	88.4	77.8	66.2	56.5	(25)	
		MCWB	45.6	50.7	56.9	62.6	69.5	74.3	76.0	75.0	71.4	64.0	55.8	48.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.5	62.4	66.6	71.0	76.3	80.3	80.6	80.0	77.4	73.6	66.8	63.0	(27)	
		MCDB	65.6	70.1	78.0	82.7	86.9	92.0	94.0	94.8	91.6	84.4	74.8	68.5	(28)	
	2%	WB	54.4	59.1	64.1	68.4	74.3	78.3	78.9	78.3	75.6	71.5	63.3	57.2	(29)	
		MCDB	62.8	67.1	73.9	79.1	84.8	90.9	93.4	94.0	89.4	80.8	71.4	62.2	(30)	
	5%	WB	50.0	55.4	61.7	66.9	72.8	77.0	77.9	77.3	74.4	69.3	60.5	52.5	(31)	
		MCDB	60.1	64.3	70.8	76.8	83.2	89.0	92.3	92.7	87.4	77.5	68.0	58.9	(32)	
	10%	WB	46.2	50.6	58.7	64.8	71.0	75.5	77.0	76.0	73.2	66.8	57.8	48.0	(33)	
		MCDB	54.5	60.2	67.7	74.0	80.9	86.6	91.1	90.4	85.0	74.7	64.8	56.0	(34)	
(35) Mean Daily Temperature Range	MDBR		22.7	22.6	23.3	23.2	21.4	20.6	22.1	22.9	23.4	24.0	23.5	21.5	(35)	
	5% DB	MCDBR	31.8	30.6	28.4	27.7	24.2	23.7	26.1	27.4	25.9	27.9	28.9	29.0	(36)	
		MCWBR	18.9	17.9	15.1	14.2	10.8	8.9	7.7	8.0	9.7	13.2	15.8	18.5	(37)	
	5% WB	MCDBR	28.7	26.4	22.2	23.2	20.5	20.1	22.0	22.6	21.8	20.9	23.5	24.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.295	0.301	0.340	0.380	0.417	0.427	0.432	0.433	0.391	0.346	0.312	0.289	(40)	
	taud		2.498	2.495	2.414	2.319	2.288	2.294	2.320	2.258	2.361	2.473	2.507	2.526	(41)	
	Ebn at Noon		288	299	294	286	276	272	270	267	274	279	279	283	(42)	
	Edh at Noon		26	29	35	40	42	42	41	42	36	30	25	23	(43)	
(44) All-Sky Solar Radiation	RadAvg		849	1049	1369	1704	1872	2137	2139	1934	1609	1210	901	717	(44)	
	RadStd		84	157	88	118	188	189	118	133	148	112	77		(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	-2.75	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page